Industrial Safety Analysis Report

1. Project Overview

This project analyzes industrial accident data to understand patterns, identify high-risk areas, and provide insights to improve workplace safety.

2. Objectives

- Analyze accident trends over time
- Identify high-risk industries and locations
- Understand the impact of human factors
- Detect the most critical risks
- Evaluate accident severity patterns

3. Data Description

The dataset includes:

- Date of accident
- Industry sector
- Country / location
- Worker type
- Gender
- Accident severity
- Critical risk

4. Dataset Summary

- Total Records: 416
- Years Covered: 2016–2017
- Countries: 3
- Industry Sectors: Mining, Metals, Others
- Worker Types: Employee, Third Party, Third Party (Remote)

5. Tools Used

- Excel (Pivot Tables & Dashboard)
- Python
- Pandas
- NumPy

• Data Visualization
• Jupyter Notebook
• GitHub

6. Data Cleaning

• Removed duplicate records
• Converted date column to datetime
• Standardized column names
• Handled categorical variables

7. Key Findings

Time Analysis:
Accidents appear lower in 2017 compared to 2016, but 2017 data is incomplete. Monthly analysis shows fluctuations without a clear trend.

Industry Analysis:
• Mining has the highest number of accidents
• Metals comes second
• Others has the lowest

Location Analysis:
• Country_01 has the highest number of accidents

Worker Type Analysis:
• Third-party workers have more accidents than employees
• Including remote workers increases the gap

Severity Analysis:
• Mining has the highest number of severe accidents
• After normalization: Mining ≈ 3%, Metals ≈ 1.5%

Gender Analysis:
• Males account for the vast majority of accidents
• No severe accidents recorded for females

Critical Risk Analysis:
Most Common Risks:
• Others
• Pressed
• Manual Tools

Most Severe Risks:
• Power Lock
• Falls
• Vehicles & Mobile Equipment

8. Key Insights

• High-frequency risks are not always high-severity risks
• Third-party workers are more exposed to severe accidents
• Mining remains the most critical sector, even after normalization
• Data quality issues exist due to the large 'Others' category

9. Business Impact

The analysis helps safety teams identify high-risk sectors, prioritize corrective actions, optimize training programs, and improve accident prevention strategies through data-driven decision-making.

10. Recommendations

• Improve training programs for third-party workers
• Focus safety measures on high-severity risks
• Improve risk classification to reduce the 'Others' category
• Monitor mining operations more closely
• Strengthen contractor safety management systems

11. Conclusion

This analysis highlights the importance of going beyond simple counts and using normalized metrics to better understand risk patterns and make informed decisions.

Author

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